Chapter 1
Introduction

1.1 Why Networks?

The oft-repeated statement that "we live in a connected world" perhaps best captures, in its simplicity, why networks have come to hold such interest in recent years. From on-line social networks like Facebook to the World Wide Web and the Internet itself, we are surrounded by examples of ways in which we interact with each other. Similarly, we are connected as well at the level of various human institutions (e.g., governments), processes (e.g., economies), and infrastructures (e.g., the global airline network). And, of course, humans are surely not unique in being members of various complex, inter-connected systems. Looking at the natural world around us, we see a wealth of examples of such systems, from entire eco-systems, to biological food webs, to collections of inter-acting genes or communicating neurons.

The image of a network—that is, essentially, something resembling a *net*—is a natural one to use to capture the notion of elements in a system and their interconnectedness. Note, however, that the term 'network' seems to be used in a variety of ways, at various levels of formality. The *Oxford English Dictionary,* for example, defines the word *network* in its most general form simply as "a collection of interconnected things." On the other hand, frequently 'network' is used inter-changeably with the term 'graph' since, for mathematical purposes, networks are most commonly represented in a formal manner using graphs of various kinds. In an effort to emphasize the distinction between the general concept and its mathematical formalization, in this book we will use the term 'network' in its most general sense above, and—at the risk of the impression of a slight redundancy—we will sometimes for emphasis refer to a graph representing such a network as a 'network graph.'

The seeds of network-based analysis in the sciences, particularly its mathematical foundation of graph theory, are often placed in the 1735 solution of Euler to the now famous Königsberg bridge problem, in which he proved that it was impossible to walk the seven bridges of that city in such a way as to traverse each only once. Since then, particularly since the mid-1800s onward, these seeds have grown in a

E.D. Kolaczyk and G. Csárdi, *Statistical Analysis of Network Data with R*, Use R! 65,

DOI 10.1007/978-1-4939-0983-4_1,

number of key areas. For example, in mathematics the formal underpinnings were systematically laid, with König [92] cited as the first key architect. The theory of electrical circuits has always had a substantial network component, going back to work of Kirchoff, and similarly the study of molecular structure in chemistry, going back to Cayley. As the fields of operations research and computer science grew during the mid-1900s, networks were incorporated in a major fashion in problems involving transportation, allocation, and the like. And similarly during that time period, a small subset of sociologists, taking a particularly quantitative view towards the topic of social structure, began developing the use of networks in characterizing interactions within social groups.

More recently—starting, perhaps, in the early to mid-1990s—there has been an explosion of interest in networks and network-based approaches to modeling and analysis of complex systems. Much of the impetus for this growth derives from work by researchers in two particular areas of science: statistical physics and computer science. To the former can be attributed a seminal role in encouraging what has now become a pervasive emphasis across the sciences on understanding how the interacting behaviors of constituent parts of a whole system lead to collective behavior and systems-level properties or outcomes. Indeed the term *complex system* was coined by statistical physicists, and a network-based perspective has become central to the analysis of complex systems. To the latter can be attributed much of the theory and methodology for conceptualizing, storing, manipulating, and doing computations with networks and related data, particularly in ways that enable efficient handling of the often massive quantities of such data. Moreover, information networks (e.g, the World Wide Web) and related social media applications (e.g., Twitter), the development of which computer scientists have played a key role, are examples of some of the most studied of complex systems (arguably reflecting our continued fascination with studying ourselves!).

More broadly, a network-based perspective recently has been found to be useful in the study of complex systems across a diverse range of application areas. These areas include computational biology (e.g., studying systems of interacting genes, proteins, chemical compounds, or organisms), engineering (e.g., establishing how best to design and deploy a network of sensing devices), finance (e.g., studying the interplay among, say, the world's banks as part of the global economy), marketing (e.g., assessing the extent to which product adoption can be induced as a type of 'contagion'), neuroscience (e.g., exploring patterns of voltage dynamics in the brain associated with epileptic seizures), political science (e.g., studying how voting preferences in a group evolve in the face of various internal and external forces), and public health (e.g., studying the spread of infectious disease in a population, and how best to control that spread).

In general, two important contributing factors to the phenomenal growth of interest in networks are (i) an increasing tendency towards a systems-level perspective in the sciences, away from the reductionism that characterized much of the previous century, and (ii) an accompanying facility for high-throughput data collection, storage, and management. The quintessential example is perhaps that of the changes in biology over the past 10–20 years, during which the complete mapping of the human genome, a triumph of computational biology in and of itself, has now paved the way for fields like systems biology to be pursued aggressively, wherein a detailed understanding is sought of how the components of the human body, at the genetic level and higher, work together.

Ultimately, the study of systems such as those just described is accompanied by measurement and, accordingly, the need for statistical analysis. The focus of this book is on how to use tools in R to do statistical analysis of *network data*. More specifically, we aim to present tools for performing what are arguably a core set of analyses of measurements that are either of or from a system conceptualized as a network.

1.2 Types of Network Analysis

Network data are collected daily in a host of different areas. Each area, naturally, has its own unique questions and problems under study. Nevertheless, from a statistical perspective, there is a methodological foundation that has emerged, composed of tasks and tools that are each common to some non-trivial subset of research areas involved with network science. Furthermore, it is possible—and indeed quite useful—to categorize many of the various tasks faced in the analysis of network data across different domains according to a statistical taxonomy. It is along the lines of such a taxonomy that this book is organized, progressing from descriptive methods to methods of modeling and inference, with the latter conveniently separated into two sub-areas, corresponding to the modeling and inference of networks themselves versus processes on networks. We illustrate with some examples.

1.2.1 Visualizing and Characterizing Networks

Descriptive analysis of data (i.e., as opposed to statistical modeling and inference) typically is one of the first topics encountered in a standard introductory course in statistics. Similarly, the visualization and numerical characterization of a network usually is one of the first steps in network analysis. Indeed, in practice, descriptive analyses arguably constitute the majority of network analyses published.

Consider the network in Fig. 1.1. Shown is a visualization[1] of part of a dataset on collaborative working relationships among members of a New England law firm, collected by Lazega [98]. These data were collected for the purpose of studying cooperation among social actors in an organization, through the exchange of various types of resources among them. The organization observed was a law firm, consisting of over 70 lawyers (roughly half partners and the other half associates) in three offices located in three different cities. Relational data reflecting resource exchange

[1] The R code for generating this visualization is provided in Chap. 3.

were collected, and additional attribute information was recorded for each lawyer, including type of practice, gender, and seniority. See Lazega and Pattison [99] for additional details.

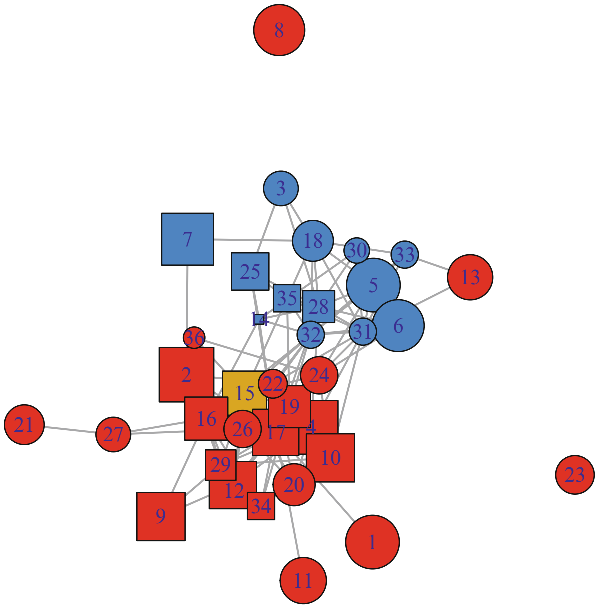

Fig. 1.1 Visualization of Lazega's network of collaborative working relationships among lawyers. *Vertices* represent partners and are labeled according to their seniority (with 1 being most senior). Vertex colors (i.e., *red*, *blue*, and *yellow*) indicate three different office locations, while vertex shape corresponds to the type of practice [i.e., litigation (*circle*) and corporate (*square*)]. Vertex area is proportional to number of years with the law firm. *Edges* indicate collaboration between partners. There are three female partners (i.e., those with seniority labels 27, 29, and 34); the rest are male

The visual summary in Fig. 1.1 manages to combine a number of important aspects of these data in one diagram. A graph is used to represent the network, with vertices corresponding to lawyers, and edges, to collaboration between pairs of lawyers. In addition, differences in vertex color, shape, size, and label are used to indicate office location, type of practice, years with the law firm, and seniority. However, this is far from the only manner in which to visualize these data. Deciding how best to do so is itself both an art and science. Furthermore, visualization is aided here by the fact that the network of lawyers is so small. Suppose instead our interest lay in a large on-line social network, such as the Facebook network of people that have 'friended' each other. With well over 1 billion active users reported as of this writing, it is impossible to display this network in an analogous manner, with every individual user and their friendships evident. A similar statement often can be made in biology, for example, such as for networks of proteins and their affinity for binding or of genes and their regulatory relationships. As a result, the visualization of large networks is a separate challenge of its own. We will look at tools for network visualization in Chap. 3.

Characterization of network data through numerical summaries is another important aspect of descriptive analysis for networks. However, unlike in an introductory statistics course, where the types of summaries encountered typically are just simple measures of center (e.g., mean, median, etc.) and dispersion (e.g., standard deviation, range, etc.) for a set of real numbers, summary measures for networks necessarily seek to capture characteristics of a graph. For example, for the lawyer data in Fig. 1.1, it is natural to ask to what extent two lawyers that both work with a third lawyer are likely to work with each other as well. This notion corresponds to the social network concept of *transitivity* and can be captured numerically through an enumeration of the proportion of vertex triples that form triangles (i.e., all three vertex pairs are connected by edges), typically summarized in a so-called clustering coefficient. In this case, the relevant characterization is an explicit summary of network structure. On the other hand, given the two types of lawyers represented in these data (i.e., corporate and litigation), it is also natural to ask to what extent lawyers of each type collaborate with those of same versus different types. This notion corresponds to another social network concept—that of *assortativity*—and can be quantified by a type of correlation statistic (the so-called assortativity coefficient), in which labels of connected pairs of vertices are compared. In this case, the focus is on an attribute associated with network vertices (i.e., lawyer practice) and the network structure plays a comparatively more implicit role.

There are a host of network summaries available, with more being defined regularly. A variety of such measures are covered in Chap. 4. These range from characterization of properties associated with individual vertices or edges to properties of subgraphs to properties of the graph as a whole. The trick to employing network characterizations in a useful fashion generally is in matching the question(s) of interest in an underlying complex system with an appropriate summary measure(s) of the corresponding network.

1.2.2 Network Modeling and Inference

Beyond asking what an observed network looks like and characterizing its structure, at a more fundamental level we may be interested in understanding how it may have arisen. That is, we can conceive of the network as having resulted from some underlying processes associated with the complex system of interest to us and ask what are the essential aspects of these processes. In addition, the actual manner in which the network was obtained, i.e., the corresponding measurement and construction process, may well be important to take into consideration. Such concerns provide the impetus for network modeling and associated tools of statistical inference.

Network modeling has received much attention. Broadly speaking, there are two classes of network models: mathematical and statistical. By ‘mathematical models' we mean models specified through (typically) simple probabilistic rules for generating a network, where often the rules are defined in an attempt to capture a particular mechanism or principle (e.g., 'the rich get richer'). In contrast, by 'statistical models' we mean models (often probabilistic as well) specified at least in part with the intention that they be fit to observed data—allowing, for example, the evaluation of the explanatory power of certain variables on edge formation in the network—and that the fit be effected and assessed using formal principles of statistical inference. While certainly there is some overlap between these two classes of models, the relevant literatures nevertheless are largely distinct.

The simplest example of a mathematical network model is one in which edges are assigned randomly to pairs of vertices based on the result of a collection of independent and identically distributed coin tosses—one toss for each vertex pair. Corresponding to a variant of the famous Erdős–Rényi formulation of a random graph, this model has been studied extensively since the 1960s. Its strength is in the fact not only that its properties are so well understood (e.g., in terms of how, for example, cohesive structure emerges as a function of the probability of an edge) but also in the role it plays as a standard against which to compare other, more complicated models.

From the statistical perspective, however, such mathematical models generally are too simple to be a good match to real network data. Nevertheless, they are not only useful in allowing for formal insight to be gained into how specific mechanisms of edge formation may affect network structure, but they also are used commonly in defining null classes of networks against which to assess the 'significance' of structural characteristics found in an observed network. For example, in Fig. 1.1, the clustering coefficient for the network of lawyer collaboration turns out to be just slightly less than 0.40. Is this value large or not? It is difficult to say without a frame of reference. One way of imposing such a frame of reference is to define a simple but comparable class of networks (e.g., all graphs with the same numbers of vertices and edges) and look at the distribution of the clustering coefficient across all such graphs. In the spirit of a permutation test and similar data re-use methods, examining where our observed value of 0.40 falls within this distribution gives us a sense of how unusual it is in comparison to values for networks across the specified class. This approach has been used not only for examining clustering in social networks but also, for example, in identifying recurring sub-graph patterns (aka 'motifs') in gene regulatory networks and functional connectivity networks in neuroscience. We will explore the use of mathematical network models for such purposes in Chap. 5.

There are a variety of statistical network models that have been offered in the literature, many of which parallel well-known model classes in classical statistics. For example, exponential random graph models are analogous to generalized linear models, being based on an exponential family form. Similarly, latent network models, in specifying that edges may arise at least in part from an unmeasured (and possibly unknown) variable(s), directly parallel the use of latent variables in hierarchical modeling. Finally, stochastic block models may be viewed as a form of mixture model. Nevertheless, importantly, the specification of such models and their fitting typically are decidedly less standard, given the usually high-dimensional and dependent nature of the data. Such models have been used in a variety of settings, from the modeling of collaborations like that in Lazega's lawyer network to the prediction of protein pairs likely to have an affinity for physically binding. We will look at examples of the use of each of these classes of models in Chap. 6.

1.2.3 Network Processes

As objects for representing interactions among elements of a complex system, network graphs are frequently the primary focus of network analysis. On the other hand, in many contexts it is actually some quantity (or attribute) associated with each of the elements in the system that ultimately is of most interest. Nevertheless, in such settings it often is not unreasonable to expect that this quantity be influenced in an important manner by the interactions among the elements, and hence the network graph may still be relevant for modeling and analysis. More formally, we can picture a stochastic process as 'living' on the network and indexed by the vertices in the network. A variety of questions regarding such processes can be interpreted as problems of prediction of either static or dynamic network processes.

Returning to Fig. 1.1, for example, suppose that we do not know the practice of a particular lawyer. It seems plausible to suspect that lawyers collaborate more frequently with other lawyers in the same legal practice. If so, knowledge of collaboration may be useful in predicting practice. That is, for our lawyer of unknown practice, we may be able to predict that practice with some accuracy if we know (i) the vertices that are neighbors of that lawyer in the network graph, and (ii) the practice of those neighbors.

While in fact this information is known for all lawyers in our data set, in other contexts we generally are not so fortunate. For example, in on-line social networks like Facebook, where users can choose privacy settings of various levels of severity, it can be of interest (e.g., for marketing purposes) to predict user attributes (e.g., perhaps indicative of consumer preferences) based on that of 'friends'. Similarly, in biology, traditionally the functional role of proteins (e.g., in the context of communication within the cell or in controlling cell metabolism) has been established through labor-intensive experimental techniques. Because proteins that work together to effect certain functions often have a higher affinity for physically binding to each other, the clusters of highly connected proteins in protein–protein interaction networks can be exploited to make computational predictions of protein function, by propagating information on proteins of known function to their neighbors of unknown function in the graph. We will encounter methods for making such predictions of static network processes in Chap. 8.

Ultimately, many (most?) of the systems studied from a network-based perspective are intrinsically dynamic in nature. Not surprisingly, therefore, many processes defined on networks are more accurately thought of as dynamic, rather than static, processes. Consider, for example, the context of public health and disease control. Understanding the spread of a disease (e.g., the H1N1 flu virus) through a population can be modeled as the diffusion of a binary dynamic process (indicating infected or not infected) through a network graph, in which vertices represent individuals, and edges, contact between individuals. Mathematical modeling (using both deterministic and stochastic models) arguably is still the primary tool for modeling such processes, but network-based statistical models gradually are seeing increased use, particularly as better and more extensive data on contact networks becomes available. Another setting in which analogous ideas and models are used—and wherein it is possible to collected substantially more data—is in modeling the adoption of new products among consumers (e.g., toward predicting the early adopters of, say, the next iPhone released). We look briefly at techniques in this emerging area of modeling and prediction of dynamic network processes in Chap. 8 as well.

In a related direction, we will look at statistical methods for analyzing network *flows* in Chap. 9. Referring to the movement of something—materials, people, or commodities, for example—from origin to destination, flows are a special type of dynamic process fundamental to transportation networks (e.g., airlines moving people) and communication networks (e.g., the Internet moving packets of information), among others.

Finally, in Chap. 10, we will look briefly at the emerging area of dynamic network analysis, wherein the network, the process(es) on the network, or indeed both, are expected to be evolving in time.

1.3 Why Use R for Network Analysis?

Various tools are available for network analysis. Some of these are standalone programs, like the classic Pajek tool or the newer Gephi, while others are embedded into a programming environment and are essentially used as a programming library. Some examples of the latter are NetworkX in Python and igraph in R.

R is rapidly becoming the de facto standard of statistical research, if this has not happened already. The majority of new statistical developments are immediately available in R and no other programming languages or software packages. While network analysis is an interdisciplinary field, it is not an exception from this trend. Several R extension packages implement one or more network analysis algorithms or provide general tools for manipulating network data and implement network algorithms. R supports high quality graphics and virtually any common graphical file format. R is extensible with currently over 5,000 extension packages, and this number is growing.

Being a complete programming language, R offers great flexibility for network research. New network analysis algorithms can be prototyped rapidly by building on the existing network science extension packages, the most commonly used one of which is the **igraph** package. In addition to implementations of classic and recently published methods of network analysis, **igraph** provides tools to import, manipulate and visualize graphs, and can be used as a platform for new algorithms. It is currently used in over a hundred other R packages and will be featured often in this book.

1.4 About This Book

Our goal in writing this book is to provide an easily accessible introduction to the statistical analysis of network data, by way of the R programming language. The book has been written at a level aimed at graduate students and researchers in quantitative disciplines engaged in the statistical analysis of network data, although advanced undergraduates already comfortable with R should find the book fairly accessible as well. At present, therefore, we anticipate the book being of interest to readers in statistics, of course, but also in areas like computational biology, computer science and machine learning, economics, neuroscience, quantitative finance, signal processing, statistical physics, and the quantitative social sciences.

The material in this book is organized to flow from descriptive statistical methods to topics centered on modeling and inference with networks, with the latter separated into two sub-areas, corresponding first to the modeling and inference of networks themselves, and then, to processes on networks. More specifically, we begin by covering tools for the manipulation of network data in Chap. 2. The visualization and characterization of networks is then addressed in Chaps. 3 and 4, respectively. Next, the topics of mathematical and statistical network modeling are investigated, in Chaps. 5 and 6, respectively. In Chap. 7 the focus is on the special case of network modeling wherein the network topology must itself be inferred. Network processes, both static and dynamic, are addressed in Chap. 8, while network flows are featured in Chap. 9. Finally, in Chap. 10 a brief look is provided at how the topics of these earlier chapters are beginning to be extended to the context of dynamic networks.

It is not our intent to provide a comprehensive coverage here of the conceptual and technical foundations of the statistical analysis of network data. For something more of that nature, we recommend the book by Kolaczyk [91]. There are also a number of excellent treatments of network analysis more generally, written from the perspective of various other fields. These include the book by Newman [118], from the perspective of statistical physics, and the book by Jackson [81], from the perspective of economics. The book by Easley and Kleinberg [49], written at an introductory undergraduate level, provides a particularly accessible treatment of networks and social behavior. Finally, the book by Wasserman and Faust [144], although less recent than these others, is still an important resource, particularly regarding basic tools for characterization and classical network modeling, from a social network perspective.

On the other hand, neither is this book a complete manual for the various R packages encountered herein. The reader will want to consult the manuals for these packages themselves for details omitted from our coverage. In addition, it should be noted that we assume a basic familiarity with R and the base package. For general background on R, particularly for those not already familiar with the software, there are any number of resources to be had, including the tutorial by Venebles and Smith [143], the beginner's guide by Zuur et al. [152], and the book by Crawley [36].

Ultimately, we have attempted to strike a reasonable balance between the concepts and technical background, on the one hand, and software details, on the other. Accordingly, we envision the book being used, for example, by (i) statisticians looking to begin engaging in the statistical analysis of network data, whether at a research level or in conjunction with a new collaboration, and hoping to use R as a natural segue; (ii) researchers from other similarly quantitative fields (e.g., computer science, statistical physics, economics, etc.) working in the area of complex networks, who seek to get up to speed relatively quickly on how to do statistical analyses (both familiar and unfamiliar) of network data in R, and (iii) practitioners in applied areas wishing to get a foothold into how to do a specific type of analysis relevant to a particular application of interest.

1.5 About the R code

The R code in this book requires a number of data sets and R packages. We have collected the data sets into a standalone R package called **sand** (i.e., for '*s*tatistical *a*nalysis of *n*etwork *d*ata'). If the reader aims to run the code while reading the book, she needs to install this package from CRAN.[2] Once installed, the package can then be used to also install all other R packages required in the code chunks of the book:

```
#1.1
> install.packages("sand")
> library(sand)
> install_sand_packages()
```

To avoid the need for constant typing while reading the book, we have extracted all R code and placed it online, at http://github.com/kolaczyk/sand. We suggest that the reader open this web page to simply copy-and-paste the code chunks into an R session. All code is also available in the **sand** package. In the package manual are details on how to run the code:

```
#1.2
> ?sand
```

Note that code chunks are not independent, in that they often rely upon the results of previous chunks, within the same chapter. On the other hand, code in separate chapters is meant to be run separately—in fact, restarting R at the beginning of each chapter is the best way to ensure a clean state to start from. For example, to run the code in Chap. 3, it is *not* required (nor recommended) to first run all code in Chaps. 1 and 2, but it *is* prudent to run any given code block in Chap. 3 only after having run all other relevant code blocks earlier in the chapter.

In writing this book, we have sought to strike an optimal balance between reproducibility and exposition. It should be possible, for example, for the reader to exactly reproduce all numerical results presented herein (i.e., summary statistics, estimates, etc.). In order to ensure this capability, we have set random seeds where necessary (e.g., prior to running methods relying on Markov chain Monte Carlo). The reader adapting our code for his own purposes will likely, of course, want to use his own choice of seeds. In contrast, however, we have compromised when it comes to

[2] The Comprehensive R Archive Network (CRAN) is a worldwide network of ftp and web servers from which R code and related materials may be downloaded. See http://cran.r-project.org.

visualizations. In particular, while the necessary code has been supplied to repeat all visualizations shown in this book, in most cases we have not attempted to wrap the main code with all necessary supplemental code (e.g., random seeds, margin settings, space between subplots, etc.) required for exact reproducibility, which would have come at what we felt was an unacceptable cost of reduction in readability and general aesthetics.

Given the fact that the R environment in general, and the R packages used in this book more specifically, can be expected to continue to evolve over time, we anticipate that we will need to update the online version of the code (and the version in the **sand** package) in the future, to make it work with newer versions of R and with updated packages. Therefore, if the reader finds that the code here in the book appears to fail, the online version should be most current and working.

On a related note, as can be seen above, the code chunks in the book are numbered, by chapter and within each chapter. This convention was adopted mainly to ensure ease in referencing code chunks from the website and vice versa. Even if we update the code on the web site, our intention is that the chunk numbers will stay the same.

In addition to having the most current version of the code chunks, the web page also includes an issue tracker, for reporting mistakes in the text or the code, and for general discussion.